

6(a)

name of gene	type of gene	name of protein produced	function of protein produced
<i>lacI</i>	regulatory ;	repressor protein	binds to operator to prevent RNA polymerase binding to promoter
<i>lacZ</i>	structural	β -galactosidase	hydrolyses / breaks down / converts, lactose to glucose and galactose ; A converts lactose to allolactose
<i>lacY ;</i>	structural	lactose permease	membrane transport protein to increase cellular uptake of lactose
<i>lacA</i>	structural	transacetylase	acetylates galactoside sugar residues to detoxify the cell

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Question	Answer	Marks
6(b)	ora for animal insulin any five from: 1 large scale supply / supply can match demand ; 2 no / less, chance of immune / allergic response or fewer side effects ; 3 fast to act / smaller dose needed ; 4 not / less, likely for (insulin) tolerance to occur ; 5 no / less, risk of, infection / disease ; 6 no / fewer, ethical / religious, considerations or suitable for, vegans / vegetarians ; 7 cheaper to produce / can control time of production / easier to purify ;	5